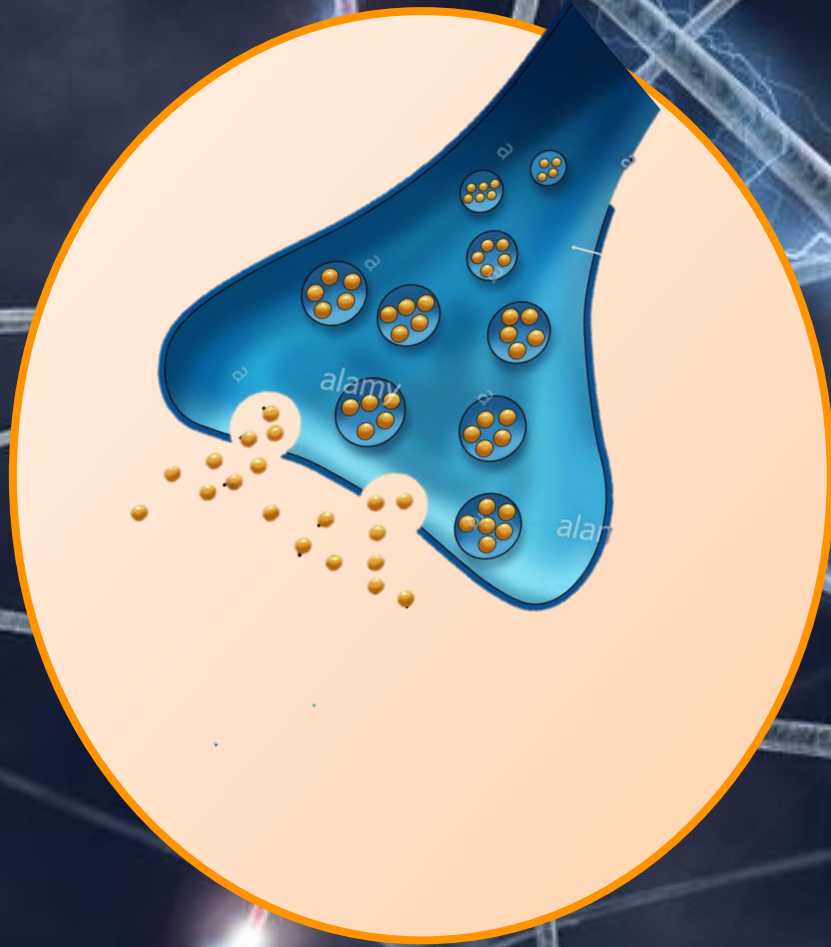
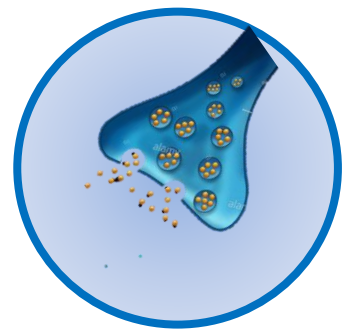




Adrenergic Agonists

Pharmacology department
Faculty of Medicine
King Abdulaziz University





Objectives:



- Give examples of a prototype drugs acting as pure alpha agonist, a pure beta agonist, and a mixed alpha and beta agonist
- Explain information related to direct-acting adrenoceptor agonists and indirect-acting adrenergic receptor agonists
- Relate the pharmacology of the autonomic nervous system to the "treatment of certain diseases" that are relevant

Central Nervous System

- Brain and spinal cord
- Integrative control centre

Peripheral Nervous System

- Peripheral nerves (cranial and spinal)
- Communication between CNS and body

Autonomic Nervous System

- Controls involuntary responses

Somatic Nervous System

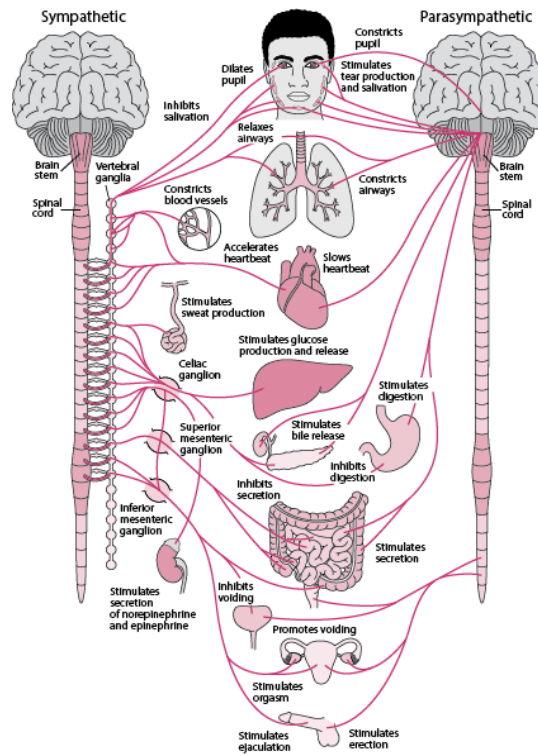
- Controls voluntary movement

Sympathetic Division

- Mobilises body systems
- 'Flight or fight' responses

Parasympathetic Division

- Conserves energy
- 'Rest and digest' responses



Speed Pharmacology

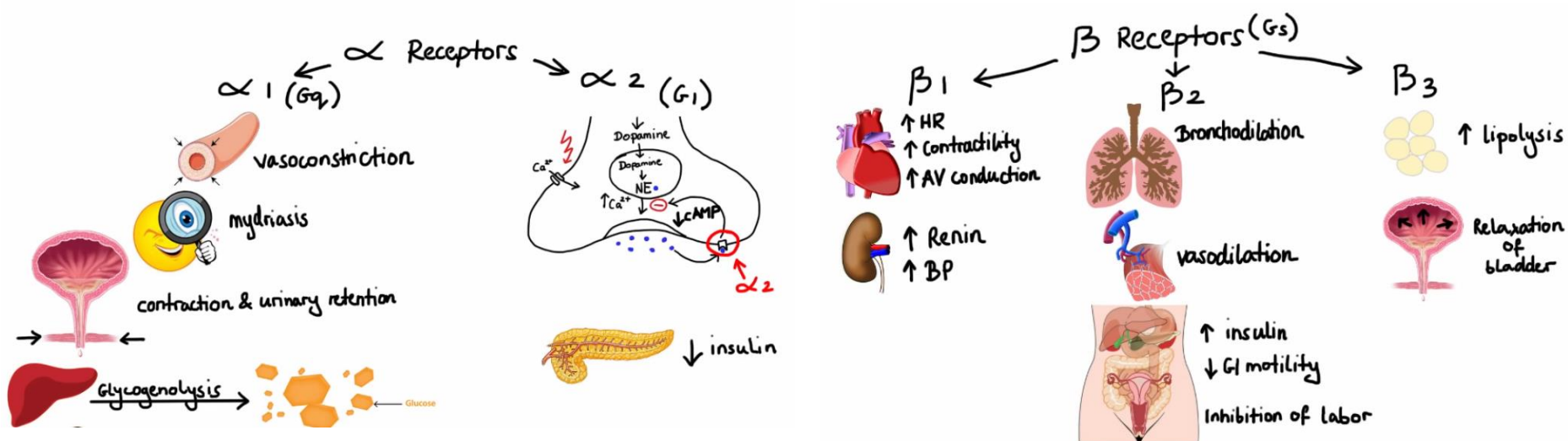
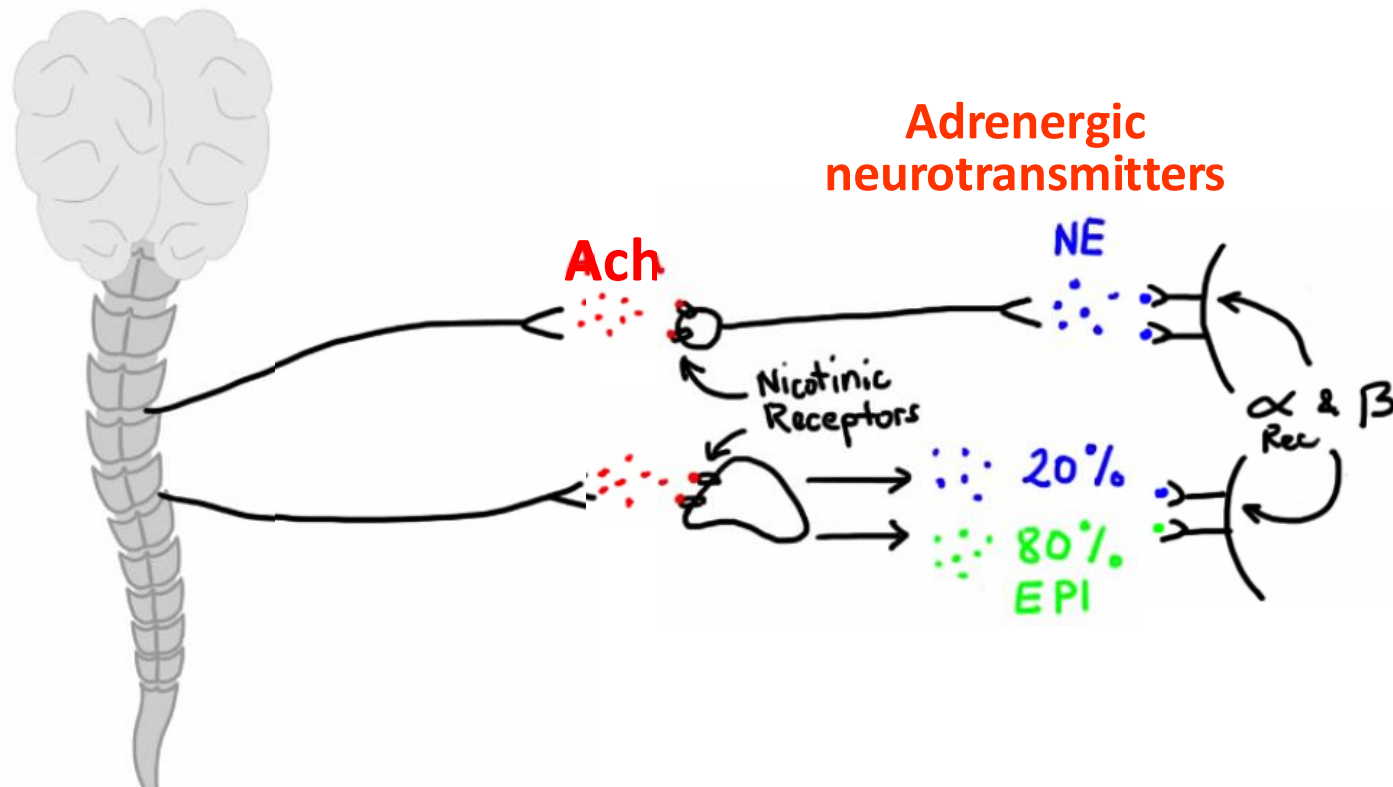
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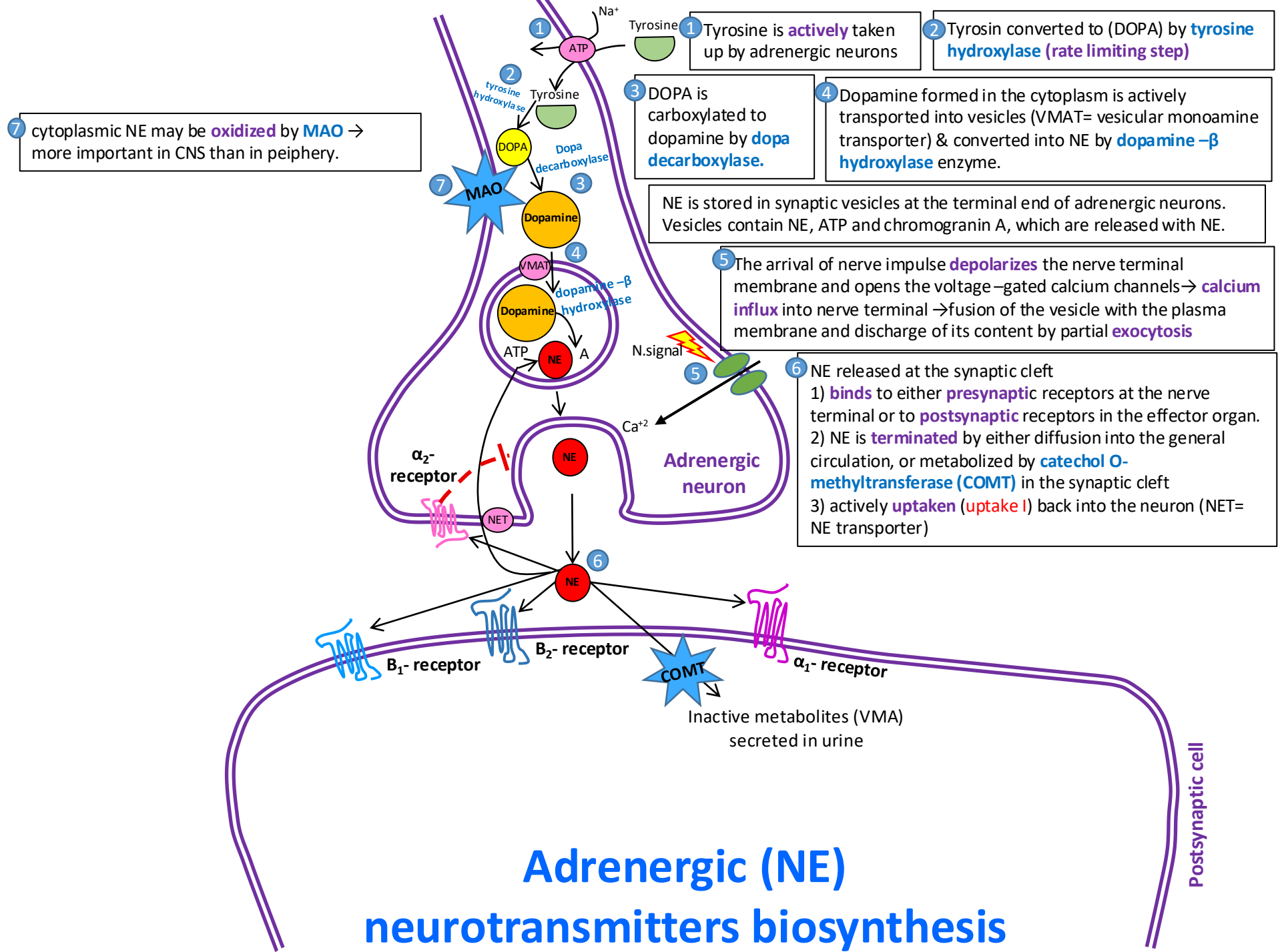


https://www.youtube.com/watch?v=qeXP_ricT4s



Adrenergic neurotransmitters





α -Adrenergic Receptors

	Location	Effect	Clinical Uses
α_1 type	Iris Radial M. Vascular SM GIT& urinary tract SM Sphincter SM prostate & vas deferens Liver	Contraction Contraction Relaxation Contraction Contraction Glycogenolysis ↑ K⁺ release	Mydriasis Shock, Local anesthesia
α_2 type (Post-synaptic)	CNS Fat cells kidney Pancreas Platelets	↓ Central sympathetic outflow ↓ lipolysis ↓ renin release. ↓ insulin release (predominant) Aggregation	Anti-hypertension

β -Adrenergic Receptors

	Location	Effect	Clinical Uses
β_1	Heart kidney Adipose tissue	$\uparrow$ Heart rate $\uparrow$ Contractility $\uparrow$ renin release Lipolysis	Cardiac Stimulant (heart failure)
β_2	Ciliary SM Ciliary body Vessel SM URT SM GI UG SM. Skeletal M. Liver,	Relaxation $\uparrow$ Aqueous humor Vasodilatation Bronchodilation Relax Glycogenolysis & $\uparrow$ K^+ uptake	glaucoma Bronchodilator (asthma)
β_3	Adipose tissue	Lipolysis	



Adrenoceptor agonists

Classification of Adrenoceptor Agonists based on Mech. of Action



Adrenergic Agonist

Direct: acting directly on α or β receptors

Indirect: induce the release of NE

Mixed action

α agonist

β agonist

Mixed α & β

Non-selective α agonist

Selective α_1 agonist

Non-selective β agonist

selective β_1 agonist

selective β_2 agonist

NE

EP

Oxymetazoline

Phenylephrine

Methoxamine

Midodrine

Isoproterenol

Dobutamine

Albuterol

Terbutaline

Formoterol

Salmeterol

Selective α_2 agonist

Alpha-methyl dopa

Clonidine

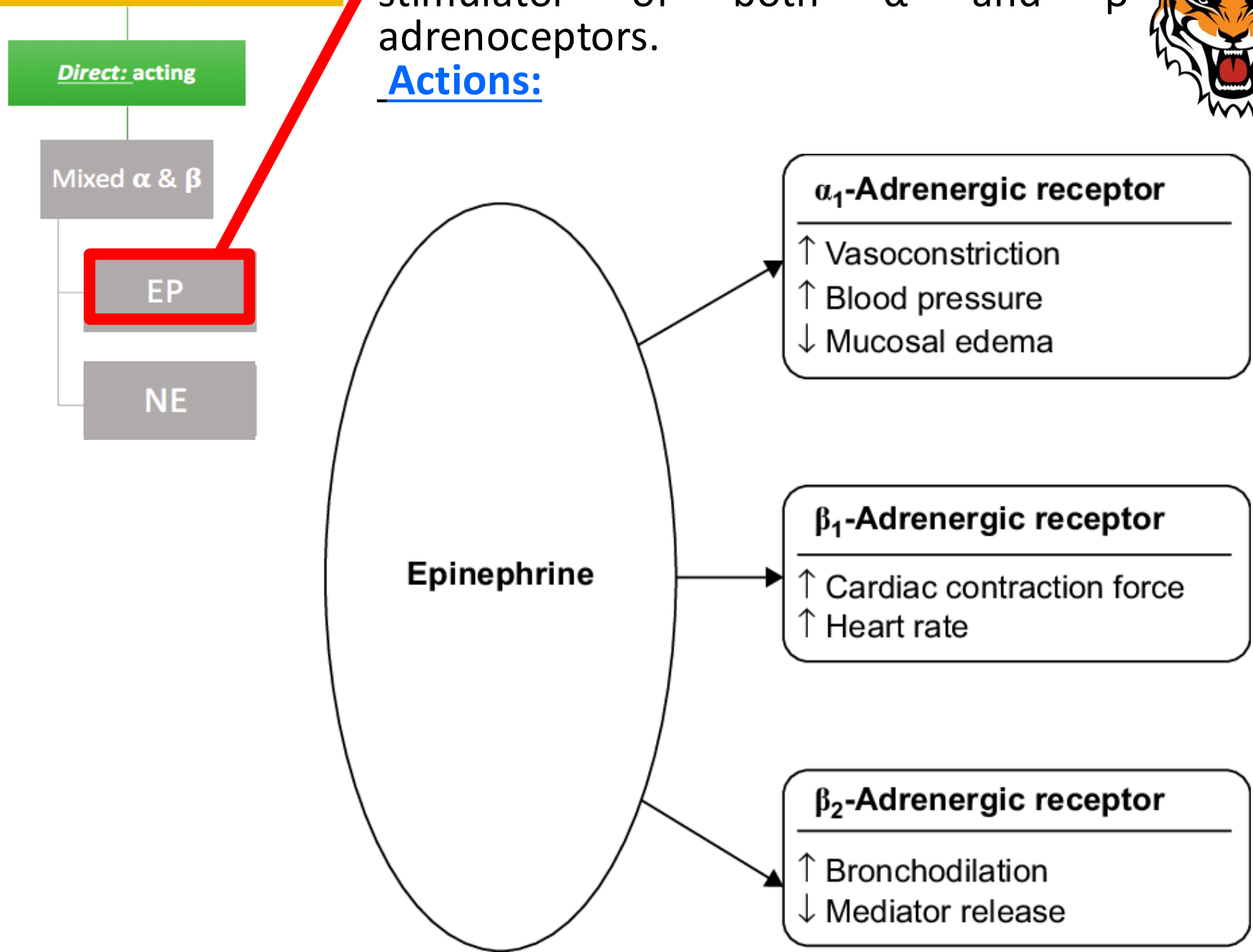
Rilmenidine

Adrenergic Agonist

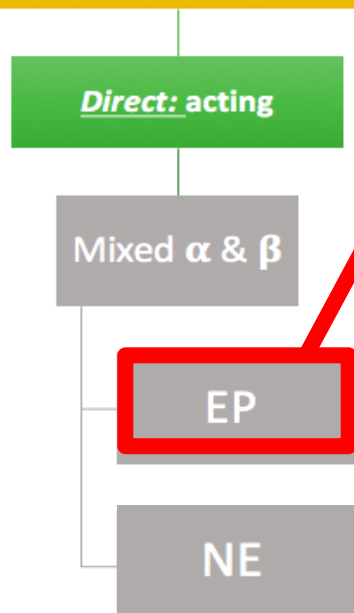
1. Epinephrine (adrenaline): Potent stimulator of both α and β adrenoceptors.



Actions:



Adrenergic Agonist



Therapeutic uses:



1- **Anaphylactic shock** (hypersensitivity /ADR) -
Bronchodilation - vasoconstriction

2- **Cardiac arrest.**

3- **Bronchial asthma** (selective ***β₂* agonists** are preferred because of minimal cardiac stimulation).

4- **Glaucoma**: 2% EP is used in chronic open angle glaucoma to decrease production of aqueous humor (VC of ciliary BV) → decreasing intraocular pressure.

5- Added to **local anesthetics** to **prolong** their action (VC → ↓ **absorption** from site of action to circulation).

6. stop **bleeding** when applied topically (in bleeding nose).

Adrenergic Agonist

Direct: acting

Mixed α & β

EP

NE



Pharmacokinetics:



Administration

Ineffective when administered **orally** as it is inactivated in GIT and liver by MAO.

- Can be given **SC** **absorption** occurs **slowly** and **prolong action** because of local VC. (avoid fingers → risk of gangrene).
- Can be given **IM**.
- **Not** given it as IV (could cause ventricular fibrillation) (only for emergency).
- **Cardiac arrest. Treatment:** Epinephrine IV (1:10,000) +CPR
- Intracardiac: Not efficient.
- Can be given by **inhalation** in Bronchial Asthma.
- Can be given **topically** on the eye.

Adrenergic Agonist



Adverse effects: (HAT)

1. **H**eadache (Throbbing Headache) & cerebral **H**emorrhage (because of rise in **BP**).
2. **A**rrhythmias (Tachycardia, palpitation).
3. **T**olerance.



Contraindications:

1. HTN
2. C.V problems (I.H.D)
3. Large dose of local anesthetic + epinephrine (sever VC); Tissue necrosis (gangrene).
4. Cardiac outflow obstruction (Hydrotrophic obstruction cardiomyopathy)
5. Hyperthyroidism (Increase T3 and T4).

T4 causes:

- 1- Increase the number of β_1 receptors (upregulation).
- 2- Increase the sensitivity of β -1 receptors
- 3- Inhibit the COMT enzyme.
- 4- Cause sympathetic over activity.



Adrenergic Agonist

2. Norepinephrine (NE):



Actions:

Potent α stimulator, weak β adrenoceptors.

Cardiovascular $\alpha 1 \gg \beta 1 > \beta 2$

$\alpha 1$

Intense Vasoconstriction
increasing peripheral
resistance, systolic and
diastolic BP

The **increased BP**
results in **reflex vagal**
stimulation which
antagonizes the
stimulant effect on
the heart rate

The end result Increase BP & reflex bradycardia.

Direct: acting

Mixed α & β

EP

NE

Adrenergic Agonist

2. Norepinephrine (NE):



Therapeutic uses:

- **Septic shock** when sever VC is needed (if BP less than 70).
- Pulmonary embolism.

Direct: acting

Mixed α & β

EP

NE

- Given as **IV infusion** only **NOT** given orally, S.C, I.M.
- IV bolus could cause tissue necrosis.

Contra-indication: With Atropine

- Any drugs increase B.P suddenly and S.B.P > 170 mm.Hg (inhibit reflex bradycardia; acute heart failure).



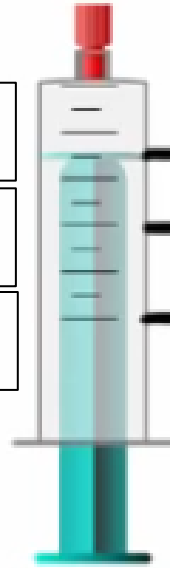
3-Dopamine: Dopamine (D) receptors, α and β adrenoceptors.

Actions: $D \rightarrow \beta_1 \rightarrow \alpha_1$

In low concentrations

In Intermediate conc

In high concentrations



D receptors

VD

β_1

+ve effect on the heart

α_1

VC

Therapeutic uses:

1. Improve myocardial function in acute HF& cardiogenic shock.
2. Chronic refractory heart failure.

Pharmacokinetics

Ineffective orally, IV use only.

- Short $T_{1/2}$ (3-5minutes).

Classification of Adrenoceptor Agonists based on Mech. of Action



Adrenergic Agonist

Direct: acting directly on α or β receptors

Indirect: induce the release of NE

Mixed action

α agonist

β agonist

Mixed α & β

Non-selective α agonist

Selective α_1 agonist

Non-selective β agonist

selective β_1 agonist

selective β_2 agonist

NE

EP

Oxymetazoline

Phenylephrine

Methoxamine

Midodrine

Isoproterenol

Dobutamine

Albuterol

Terbutaline

Formoterol

Salmeterol

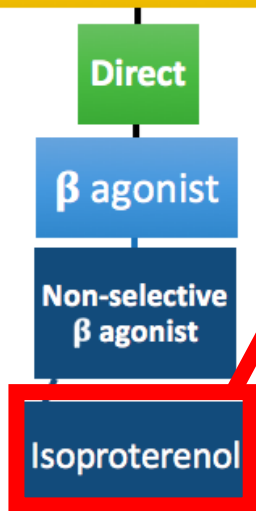
Selective α_2 agonist

Alpha-methyl dopa

Clonidine

Rilmenidine

Adrenergic Agonist



4-Isoproterenol: Potent nonselective (β_1 , β_2) adrenoceptors agonist.

Actions $\beta_1 = \beta_2$

β_1

Increased cardiac contractility and heart rate

β_2

VD of arterioles of skeletal muscle

↓ BP

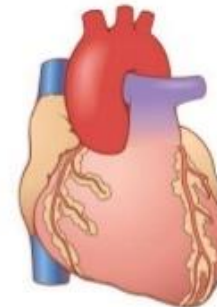
β_2

Bronchodilation

Therapeutic uses:

1. Used to stimulate heart rate (β_1) in patients w **bradycardia/ heart block**.
2. **Bronchodilator (β_2)** (rarely used now, replaced by more selective (β_2) agonists).

β_1 : Heart (stimulation)



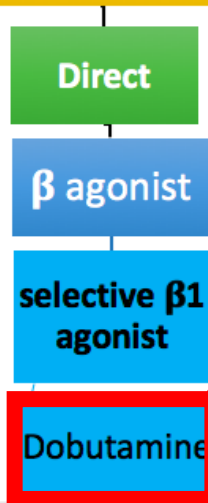
β_2 : Bronchial, uterine, and vascular smooth muscle (relaxation)



Adverse effects

Palpitations, Tachycardia, Coronary insufficiency

Adrenergic Agonist



5. Dobutamine: selective β1

Actions: β1

β1

Marked +ve inotropic and Less chronotropic.
Minimal changes in HR and systolic BP.

Dopamine	Dobutamine
Uses	
Acute HF & cardiac shock	
if there is: - hypotension (α1) and - renal impairment (D1)	if there is: - normal BP and - preserved renal function
Chronic refractory heart failure	
<u>IV infusion:</u> low dose → gradually ↑ to a maximum dose. 2 ug/kg/min → 5 ug/kg/min	<u>IV infusion:</u> low dose → gradually ↑ to maximum dose. 2.5 ug/kg/min → 10 ug/kg/min

Adrenergic Agonist

Direct

β agonist

selective β_1
agonist

Dobutamine

5. Dobutamine: selective β_1

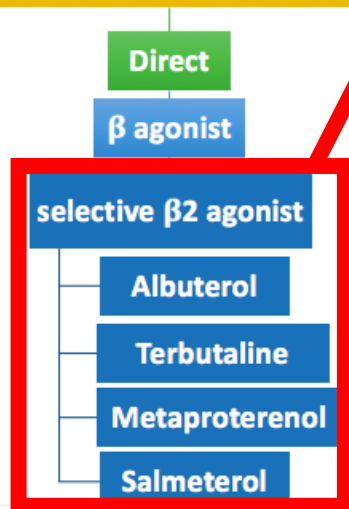


Therapeutic uses:

Sever congestive heart failure not response to digitalis.



Adrenergic Agonist



6. Selective β2 adrenoceptors agonists:

Actions: β2

Bronchodilation- **inhibits** uterine contractions



Pharmacokinetics

- **Short acting (SABA):** Metaproterenol, **albuterol** (ventolin), salbutamol and **terbutaline**

- **Long acting (LABA):** Salmeterol and formeterol.

- **Long** duration (not metabolized by MAO or COMT).

- Route of administration: oral, inhalation & parenteral

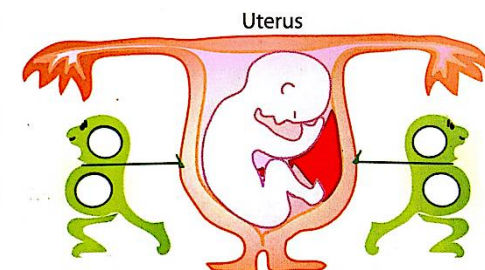
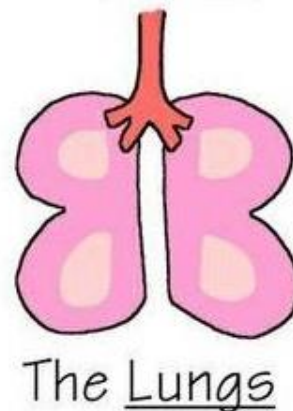
Therapeutic uses:

- **Bronchial asthma.**

- **Terbutaline** reduce uterine contraction in premature labor.

Adverse effects

1. Tachycardia and palpitations (B1).
2. Tremor.
3. Hypokalemia
4. Hyperglycemia



Classification of Adrenoceptor Agonists based on Mech. of Action



Adrenergic Agonist

Direct: acting directly on α or β receptors

Indirect: induce the release of NE

Mixed action

α agonist

β agonist

Mixed α & β

Non-selective α agonist

Selective α_1 agonist

Non-selective β agonist

selective β_1 agonist

selective β_2 agonist

NE

EP

Oxymetazoline

Phenylephrine

Methoxamine

Midodrine

Isoproterenol

Dobutamine

Albuterol

Terbutaline

Formoterol

Salmeterol

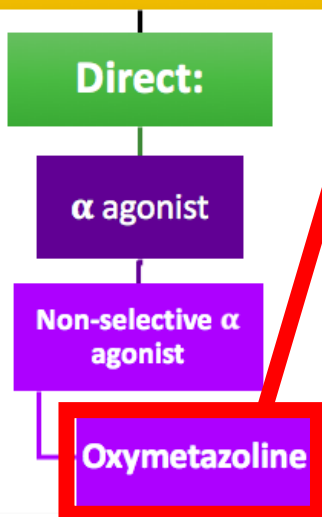
Selective α_2 agonist

Alpha-methyl dopa

Clonidine

Rilmenidine

Adrenergic Agonist



7. Non-Selective α agonists: Oxymentazoline



Action: α_1 , α_2

VC

Pharmacokinetics

- Route of administration: Local eye/ nose drops

Therapeutic uses:

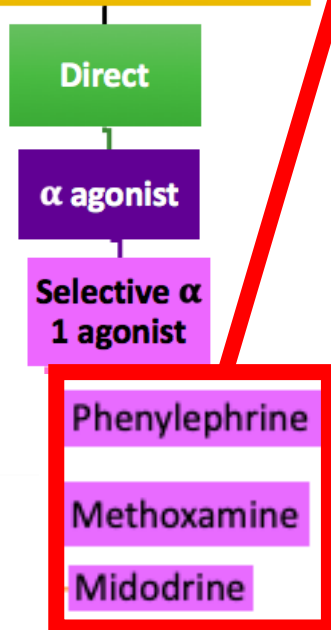
nasal and eye **decongestant**



Adverse effects

headache, sneezing, burning of the nasal mucosa and **rebound congestion**.

Adrenergic Agonist



8. Selective α_1 agonists:

Action: α_1

-VC $\rightarrow$ $\uparrow$ BP with reflex bradycardia.



Therapeutic uses:

Phenylephrine

1. **Nasal decongestant** and epistaxis.
2. **Red eye** (10% ophthalmic solution).
3. Vasopressor agent (An adjunct for use with local anesthetics).
4. Postural hypotension.

Methoxamine

It is used in hypotensive states (parenteral).

Midodrine

It is mainly used in postural hypotension

Adverse effects

1. Hypertension $\rightarrow$ reflex bradycardia.
2. Rebound nasal congestion and **atrophic rhinitis**.

Selective α_2 agonists Alpha-methyl dopa, Clonidine, Rilmenidine

CENTRALLY-ACTING SYMPATHOLYTICS

Action: α_2

acts centrally (brain) on postsynaptic α_2 adrenoceptors to *inhibit central sympathetic outflow*.

Also they may stimulate presynaptic α_2 adrenoceptors to reduce norepinephrine release.

Therapeutic uses:

1. *treatment of **hypertension***
2. alcohol and opiates **withdrawal**

Precaution

Sudden or abrupt withdrawal can result in *severe hypertension* treated by both α and β blockers.

Classification of Adrenoceptor Agonists based on Mech. of Action



Adrenergic Agonist

Direct: acting directly on α or β receptors

Indirect: induce the release of NE

Mixed action

α agonist

β agonist

Mixed α & β

Non-selective α agonist

Selective α_1 agonist

Non-selective β agonist

selective β_1 agonist

selective β_2 agonist

NE

EP

Oxymetazoline

Phenylephrine

Methoxamine

Midodrine

Isoproterenol

Dobutamine

Albuterol

Terbutaline

Formoterol

Salmeterol

Selective α_2 agonist

Alpha-methyl dopa

Clonidine

Rilmenidine

Adrenergic Agonists

B. Indirect Acting adrenergic Agonists

1. Reuptake inhibitors:
Cocaine

2. α_2 -Antagonist:
Yohimbine

3. Release stimulants

Amphetamine

Therapeutic uses: depression, hyperactivity in children, narcolepsy and appetite control.

Tyramine

NO clinical uses but it is found in fermented food.

Drug-Drug interaction: with MAOI causes hypertensive crisis

C. Mixed-action adrenergic agonist

Ephedrine & Pseudoephedrine

Therapeutic uses: Nasal congestion, and hypotension

4. Selective β_1 agonists:

Dobutamine

Therapeutic uses: refractory CHF.

6. Non-selective β agonists

Isoproterenol

Therapeutic uses: Cardiac stimulation in in AV block.

A. Direct Acting adrenergic Agonists

1. Mixed α and β agonists

Norepinephrine (α and β_1 agonists)
Therapeutic uses: shock

Epinephrine

Therapeutic uses: for Anaphylactic shock, cardiac arrest, bronchospasm, and with local anesthesia.

2. Non-selective α agonists:

Oxymetazoline

Therapeutic uses: used as nasal decongestant, and ophthalmic drops for eye redness.

3. Selective α_1 agonists:

Phenylephrine

Therapeutic uses: used as nasal decongestant, as a mydriatic agent for eye examination, for hypotension.

5. Selective β_2 agonists

Albuterol

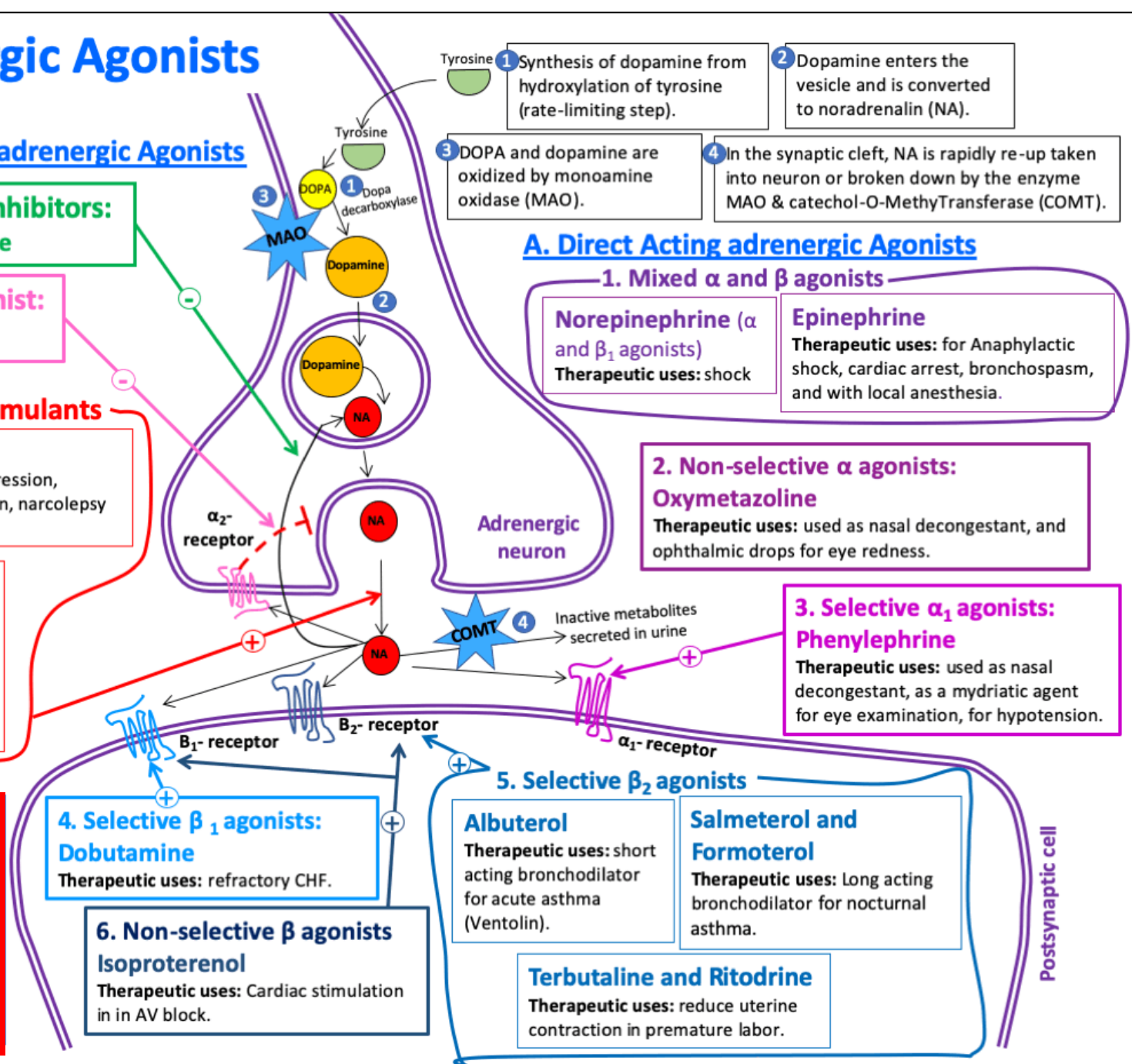
Therapeutic uses: short acting bronchodilator for acute asthma (Ventolin).

Salmeterol and Formoterol

Therapeutic uses: Long acting bronchodilator for nocturnal asthma.

Terbutaline and Ritodrine

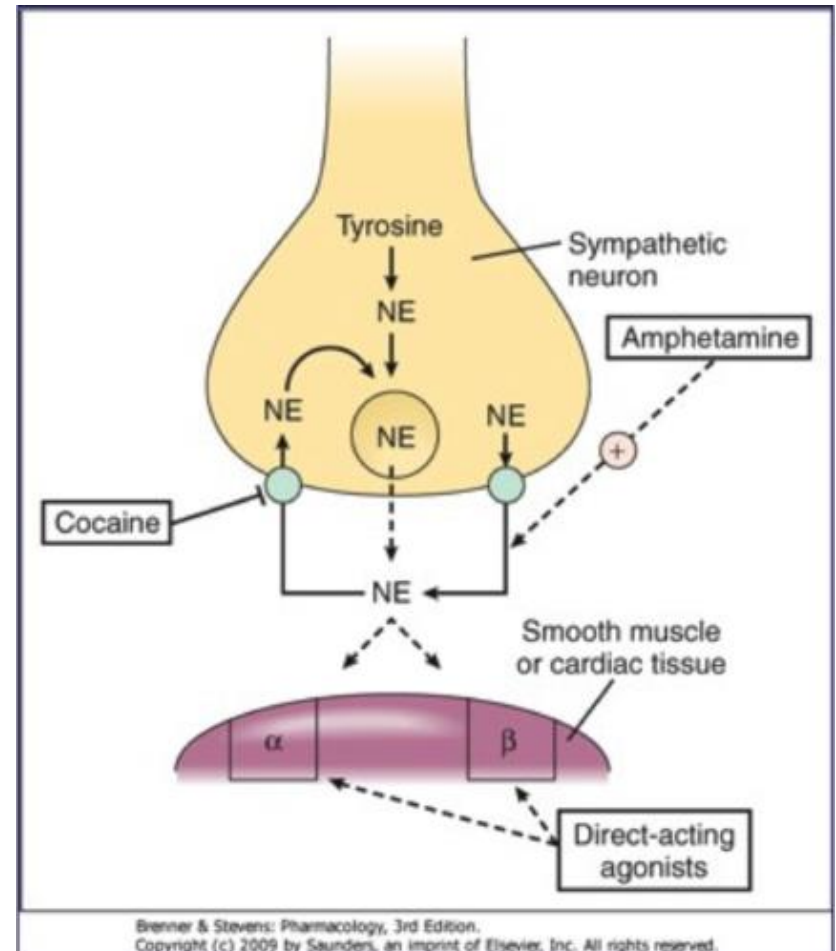
Therapeutic uses: reduce uterine contraction in premature labor.



Indirect-Acting Adrenergic Agonists

Mechanism of action

- (1) Displacement of stored catecholamines: **amphetamine**, **tyramine**.
- (2) Inhibition of reuptake of catecholamines: **cocaine**, tricyclic antidepressants.



1- Indirect-Acting Adrenergic Agonists: Amphetamines

Action by releasing NE

1. **CNS:** CNS stimulation, alertness, ↓fatigue- ↑↑mood elevation - ↓ Appetite
2. **CVS:** ↑ arterial blood pressure → reflex bradycardia.

Therapeutic uses:

1. Treatment of narcolepsy (sleep disorder).
2. The appetite-suppressing effect: (Weight reduction).
3. Attention deficit hyperactivity disorder in children.

Adverse effects

- CNS:**
- 1- Psychological dependence & Schizophrenia-like syndrome.
 - 2- Anorexia (loss of appetite) & weight loss.
 - 3- Insomnia, tremors → depression, fatigue (CA store depletion)
 - 4- Severe toxicity → convulsion, coma & cerebral hemorrhage
- CVS:** Palpitation, arrhythmia, anginal pain and hypertension.

2- Indirect-Acting Adrenergic Agonists: Tyramine

Action by releasing NE

- It is found in high concentrations in fermented foods (cheese..)
- Release of stored catecholamines.
- It metabolized by MAO in liver
- With MAO-Is  hypertensive crisis ($\uparrow$ BP, $\uparrow$ HR)

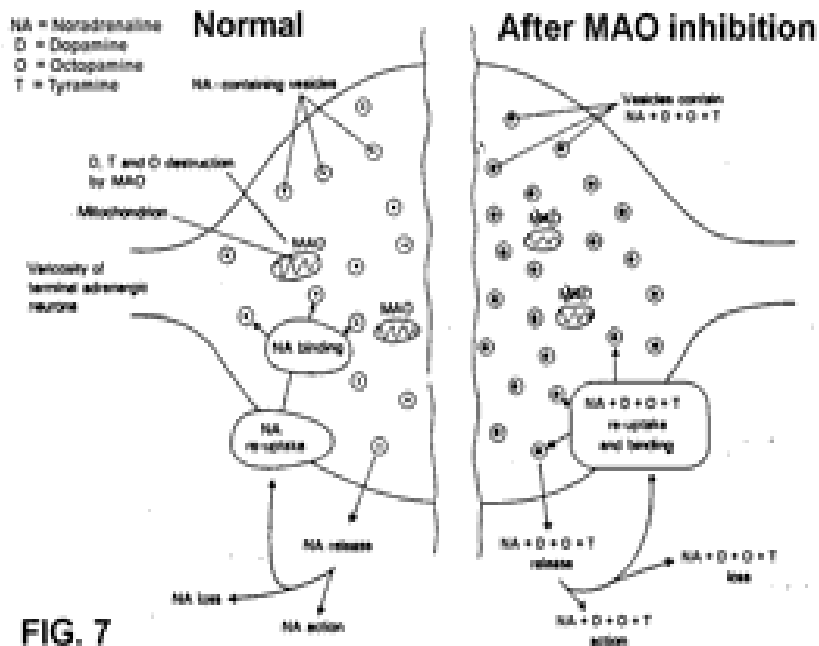


FIG. 7



3- Indirect-Acting Adrenergic Agonists: Cocaine

Action:

1. Block norepinephrine reuptake
2. Block sodium channels, as a local anesthetic.

Therapeutic uses:

A potent central nervous system stimulant.

Adverse effects

Overdose: causes tachyarrhythmias and a marked elevation of blood pressure.

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Indirect: induce the release of NE

Mixed action

α agonist

β agonist

Mixed α & β

Non-selective α agonist

Selective α_1 agonist

Non-selective β agonist

selective β_1 agonist

selective β_2 agonist

NE

EP

Oxymetazoline

Phenylephrine

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Dobutamine

Albuterol

Terbutaline

Formoterol

Salmeterol

Selective α_2 agonist

Alpha-methyl dopa

Clonidine

Rilmenidine

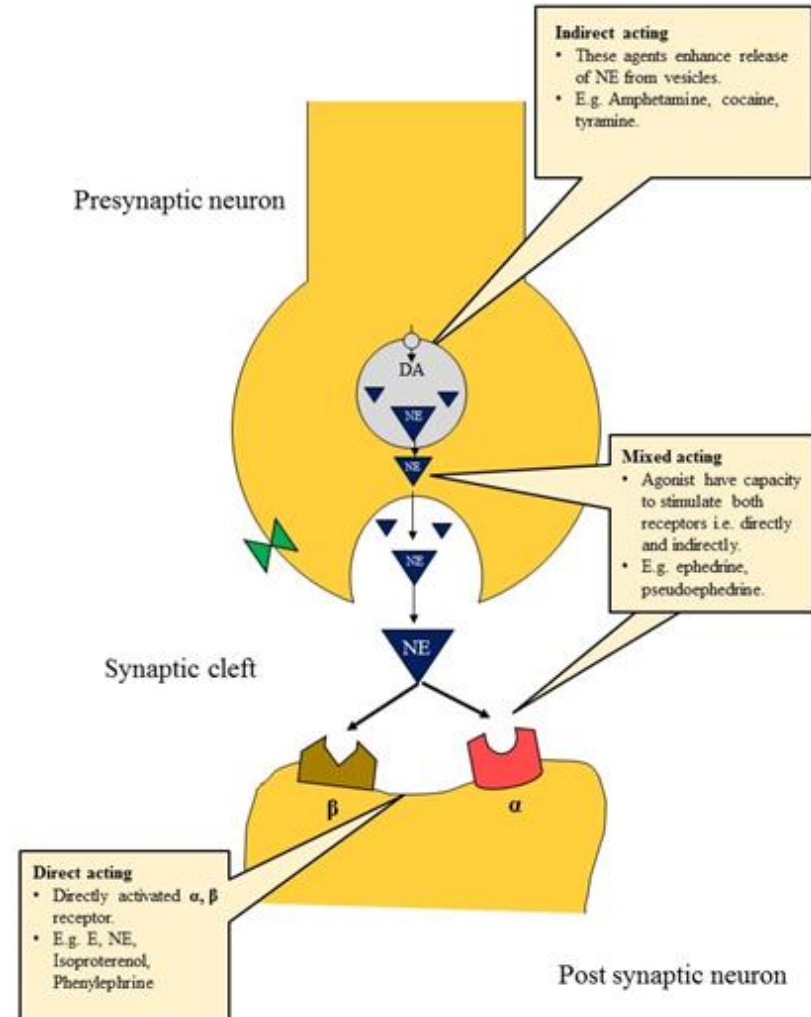
Methoxamine

Midodrine

Mixed-acting sympathomimetic

Mechanism of action

- **Directly** stimulates the receptor by binding to it.
- **AND**
- **Indirectly** stimulates the receptor by causing the release of stored neurotransmitters from the vesicles in the nerve endings.



Mixed-acting sympathomimetic: Ephedrine & Pseudoephedrine

Mechanism of action

1. **(Direct)** actions on adrenoceptors.
2. **(Indirect)** release of stored catecholamines.

Pharmacokinetics

Resistant to COMT and MAO (not catechol), Long duration of action

Therapeutic uses:

1. Topical hemostatic in epistaxis (ephedrine).
2. Nasal decongestant: ephedrine, pseudoephedrine (systemic).
3. Spinal shock (IV ephedrine).

Adverse effects

- 1- Minimal CNS (+) → insomnia, anxiety (sleepiness in children).
- 2- Minimal CVS (+) → palpitation, arrhythmia, anginal pain.
- 3- Urinary retention.

Adrenergic Agonists

B. Indirect Acting adrenergic Agonists

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Therapeutic uses: Nasal congestion, and hypotension

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Therapeutic uses: refractory CHF.

6. Non-selective β agonists
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Therapeutic uses: Cardiac stimulation in in AV block.

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5. Selective β_2 agonists

Albuterol

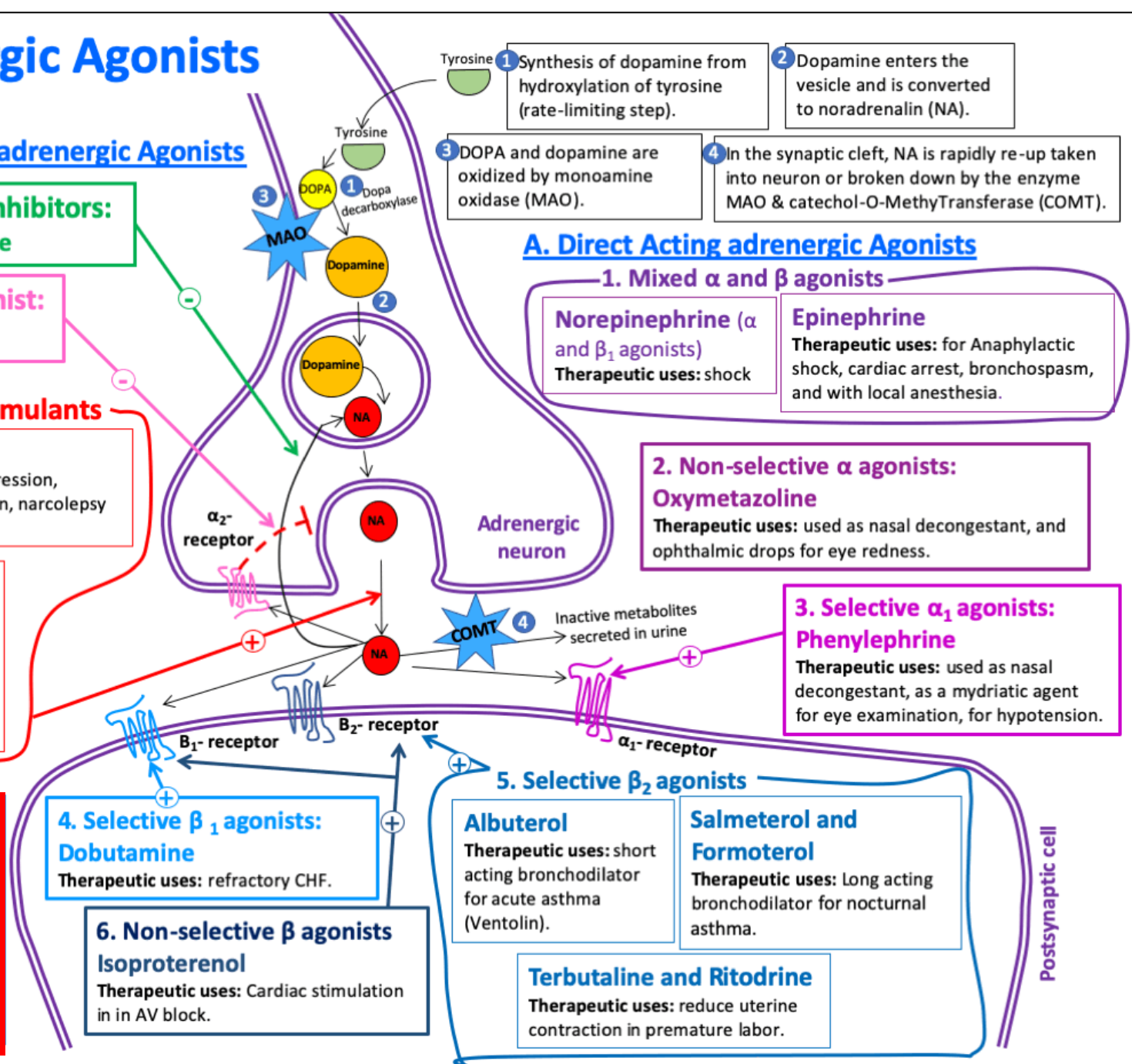
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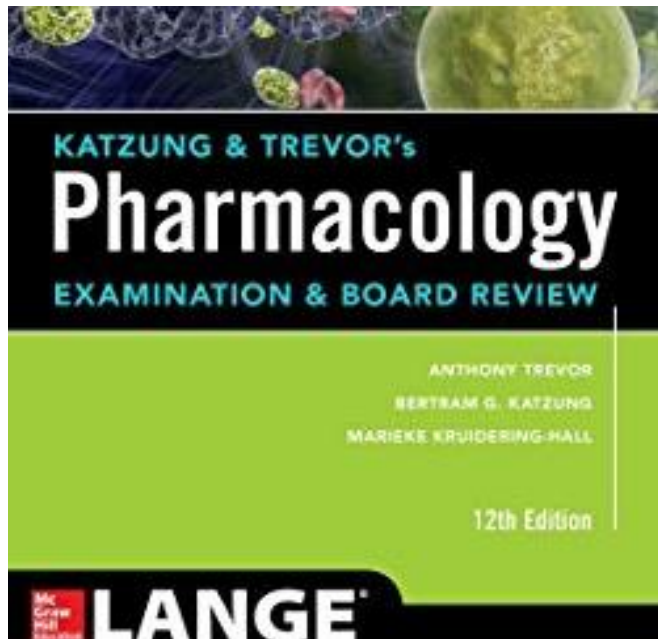
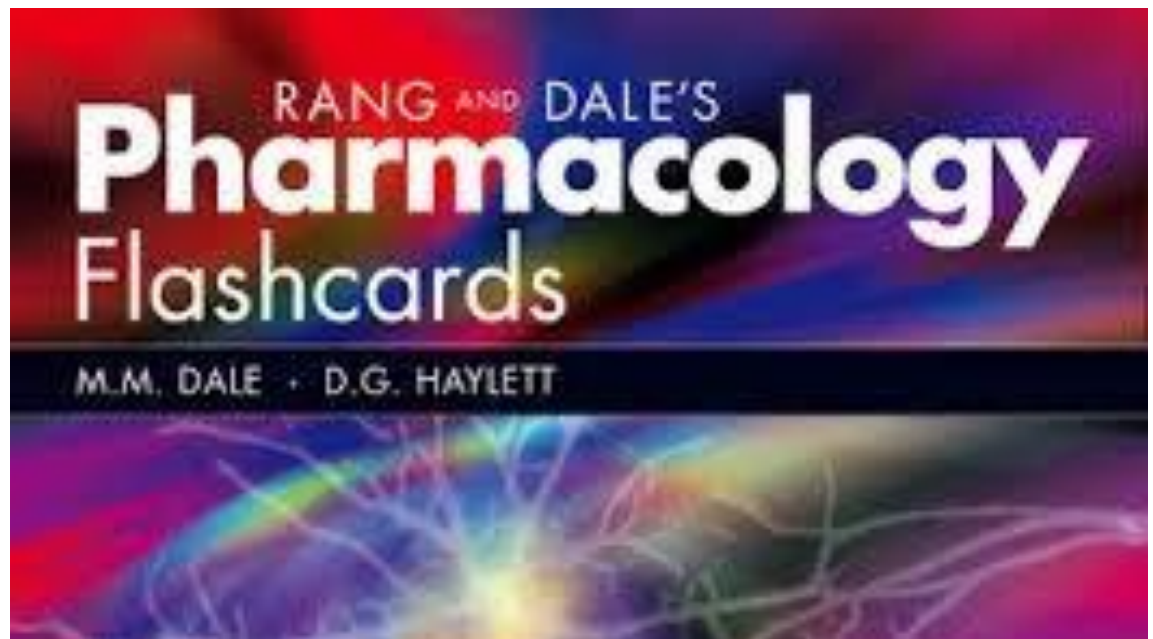
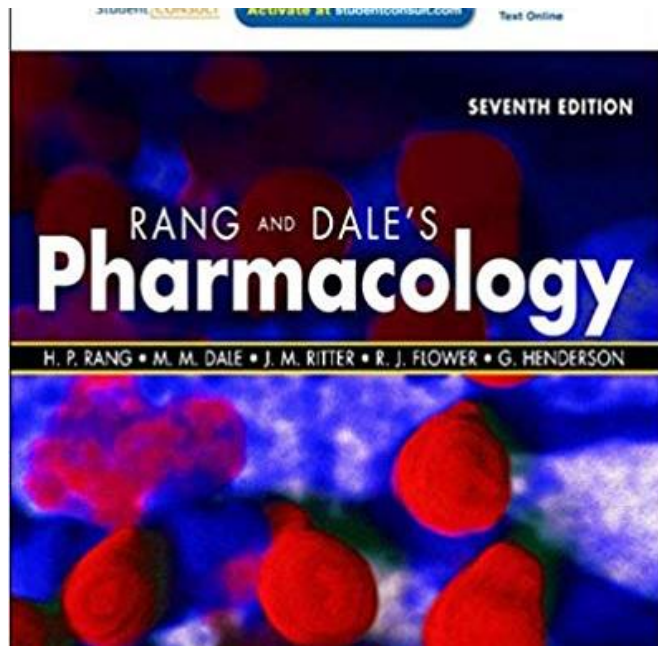
Salmeterol and Formoterol

Therapeutic uses: Long acting bronchodilator for nocturnal asthma.

Terbutaline and Ritodrine

Therapeutic uses: reduce uterine contraction in premature labor.





References



https://www.youtube.com/watch?v=s4m_cdYvRug&feature=youtu.be

Quiz:

<https://forms.gle/p5DhjfrXEde3d6P9>

